Integration and Economic Dynamics

ECON 320 - Introduction to Mathematical Economics

Muhebullah Karimzada

Fall 2025

1. Motivation: Why Integration in Economics?

So far, you may be familiar with **derivatives**:

$$\frac{dy}{dx}, \quad f'(x), \quad \frac{dQ}{dP}, \quad \text{etc.}$$

Derivatives describe *instantaneous change*. For example:

- Marginal cost is the derivative of total cost with respect to quantity.
- Marginal utility is the derivative of total utility with respect to consumption.
- Marginal product is the derivative of output with respect to an input.

However, many economic questions are about **accumulation over time or over quantity**, such as:

- Given marginal cost, what is total cost between $Q = 0$ and $Q = 100$?
- Given investment as a function of time, what will the capital stock be after 10 years?
- Given an income flow $Y(t)$, what is the present value of all future income?

To answer such questions, we need a mathematical tool that *adds up* small contributions:

$$\text{Total} = \text{sum of many tiny pieces.}$$

This is exactly what **integration** does.

2. What is an Integral? First Intuition

2.1 Area Under a Curve as a Sum

Consider a function $f(x)$ defined on the interval $[a, b]$. Visually, we can think of $f(x)$ as a curve. Suppose we want the **area under the curve** from $x = a$ to $x = b$.
A natural idea:

1. Divide $[a, b]$ into many small sub-intervals of width Δx.

2. On each small interval, approximate the area by a rectangle:

$$\text{height} \approx f(x_i), \quad \text{width} = \Delta x.$$

3. Add up all these small rectangular areas:

$$\text{Total area} \approx \sum_i f(x_i)\Delta x.$$

As we make Δx smaller and smaller (more and more rectangles), this approximation becomes better and better.

2.2 The Limit: From Sum to Integral

Formally, the **definite integral** of $f(x)$ from a to b is defined as:

$$\int_a^b f(x)\,dx = \lim_{\Delta x \to 0} \sum_i f(x_i)\Delta x.$$

This is a limit of sums. That is why we say the integral is a *continuous sum*.

Definite Integral: $\displaystyle\int_a^b f(x)\,dx$ is the limiting value of sums of the form $\sum f(x_i)\Delta x$ as $\Delta x \to 0$. It represents the *area under the curve* $y = f(x)$ from $x = a$ to $x = b$ (when $f(x) \geq 0$).

3. Relationship Between Derivatives and Integrals

It turns out that there is a deep relationship between differentiation and integration. Suppose $F(x)$ is a function such that

$$F'(x) = f(x).$$

Then we say $F(x)$ is an **antiderivative** (or **indefinite integral**) of $f(x)$.

Fundamental Theorem of Calculus (informal)

If $F'(x) = f(x)$, then:

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

In words: to compute the definite integral of $f(x)$ from a to b, find any function $F(x)$ whose derivative is $f(x)$, then take $F(b) - F(a)$.

This theorem tells us:

- Integration can be done by finding antiderivatives.
- Differentiation and integration are inverse operations.

4. Indefinite Integrals: Basic Rules (With Steps)

An **indefinite integral** has the form:

$$\int f(x)\,dx = F(x) + C,$$

where $F'(x) = f(x)$ and C is an arbitrary constant (because the derivative of a constant is zero).

4.1 Power Rule

We start from the derivative:

$$\frac{d}{dx}x^{n+1} = (n+1)x^n.$$

We want to *reverse* this. Suppose we want $\int x^n dx$.

Step 1: Guess that the antiderivative has the form Ax^{n+1}.

Step 2: Differentiate:

$$\frac{d}{dx}(Ax^{n+1}) = A(n+1)x^n.$$

Step 3: We want this to equal x^n, so set:

$$A(n+1) = 1 \quad \Rightarrow \quad A = \frac{1}{n+1}.$$

Conclusion:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad \text{for } n \neq -1.$$

4.2 Integral of $1/x$

We know

$$\frac{d}{dx} \ln|x| = \frac{1}{x}.$$

Thus:

$$\int \frac{1}{x} dx = \ln|x| + C.$$

4.3 Exponential Functions

We know:

$$\frac{d}{dx} e^{ax} = ae^{ax}.$$

To find $\int e^{ax} dx$, follow similar steps:

Step 1: Guess that the antiderivative is Be^{ax}.

Step 2: Differentiate:

$$\frac{d}{dx}(Be^{ax}) = Bae^{ax}.$$

Step 3: We want this to equal e^{ax}, so:

$$Ba = 1 \quad \Rightarrow \quad B = \frac{1}{a}.$$

Conclusion:

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C.$$

4.4 Linearity of the Integral

If k is a constant and f, g are functions:

$$\int [kf(x)]dx = k \int f(x)dx,$$

$$\int [f(x) + g(x)]dx = \int f(x)dx + \int g(x)dx.$$

We can justify this by using the linearity of differentiation and then reversing the process.

Example (step-by-step):

$$\int (3x^2 + 4x)dx.$$

Step 1: Apply linearity:

$$\int (3x^2 + 4x)dx = \int 3x^2dx + \int 4xdx.$$

Step 2: Pull constants out:

$$= 3 \int x^2dx + 4 \int xdx.$$

Step 3: Use power rule:

$$= 3\left(\frac{x^3}{3}\right) + 4\left(\frac{x^2}{2}\right) + C = x^3 + 2x^2 + C.$$

5. Definite Integrals and the Fundamental Theorem of Calculus

5.1 Definite Integrals

A **definite integral** from a to b is written:

$$\int_a^b f(x)dx.$$

This is a number (not a function). Using an antiderivative $F(x)$ with $F'(x) = f(x)$:

$$\int_a^b f(x)dx = F(b) - F(a).$$

Example:

$$\int_0^2 (3x^2 + 4x)dx.$$

We already know antiderivative:

$$\int (3x^2 + 4x)dx = x^3 + 2x^2 + C.$$

So:

$$\int_0^2 (3x^2 + 4x)dx = [x^3 + 2x^2]_0^2 = (2^3 + 2 \cdot 2^2) - (0^3 + 2 \cdot 0^2).$$

Compute step-by-step:

$$2^3 = 8, \quad 2^2 = 4, \quad 2 \cdot 4 = 8.$$

So:

$$x^3 + 2x^2 \text{ at } x = 2 \text{ is } 8 + 8 = 16,$$

and at $x = 0$:

$$0^3 + 2 \cdot 0^2 = 0.$$

Therefore:

$$\int_0^2 (3x^2 + 4x)dx = 16 - 0 = 16.$$

Economic Reading: If $3x^2+4x$ were the marginal cost, then the total cost of producing from 0 to 2 units would be 16 (plus any fixed cost).

6. Economic Applications: Static (No Time Yet)

6.1 Total Cost from Marginal Cost

Suppose marginal cost is $MC(Q)$, and fixed cost is F. Then:

$$TC(Q) = F + \int_0^Q MC(q)dq.$$

Example:

$$MC(Q) = 10 + 2Q, \quad F = 50.$$

Step 1: Compute the integral:

$$\int_0^Q (10 + 2q)dq = \int_0^Q 10\,dq + \int_0^Q 2q\,dq.$$

Step 2: Integrate separately:

$$\int_0^Q 10\,dq = 10Q, \quad \int_0^Q 2q\,dq = 2 \cdot \frac{Q^2}{2} = Q^2.$$

Step 3: Add:

$$\int_0^Q (10 + 2q)dq = 10Q + Q^2.$$

Step 4: Add fixed cost:

$$TC(Q) = 50 + 10Q + Q^2.$$

6.2 Consumer Surplus

Demand curve: $P(Q)$, market price: P^*. Quantity traded: Q^*.

$$CS = \int_0^{Q^*} P(Q)dQ - P^*Q^*.$$

This is literally the area under the demand curve above the horizontal price line.

7. Integration and Economic Dynamics: Stocks and Flows

Now we move from *static* applications to *dynamic* ones.

7.1 Flows Accumulating into Stocks

Let $I(t)$ be net investment (a **flow**), measured per unit of time.
Capital stock $K(t)$ is a **stock**.
The change in capital over a small time Δt is approximately:

$$\Delta K \approx I(t)\Delta t.$$

To get total change over $[0, t]$, we add up all tiny changes:

$$K(t) - K(0) \approx \sum I(t_i)\Delta t.$$

In the limit:

$$K(t) - K(0) = \int_0^t I(s)ds.$$

So:

$$\boxed{K(t) = K(0) + \int_0^t I(s)ds.}$$

This is the basic **stock-flow** relationship.

7.2 Example: Constant Investment

Suppose $I(t) = \bar{I}$, a constant.
Then:

$$K(t) = K(0) + \int_0^t \bar{I}\ ds = K(0) + \bar{I}t.$$

This is just a straight line: capital grows at a constant rate over time.

8. Introducing Differential Equations

A **differential equation** is an equation that involves an unknown function and its derivatives.

Example:

$$\frac{dK}{dt} = I(t).$$

This equation says: the rate of change of K with respect to time equals the investment flow. If we know $I(t)$, we can integrate to get $K(t)$:

$$K(t) = K(0) + \int_0^t I(s)ds.$$

So integration is the natural tool for solving many differential equations.

9. Example: Capital Accumulation with Depreciation

Consider:

$$\frac{dK}{dt} = I - \delta K,$$

where:

- I is *constant net investment.*
- δ is the *depreciation rate.*

This is a first-order linear differential equation.

9.1 Rewrite the Equation

We can rewrite as:

$$\frac{dK}{dt} + \delta K = I.$$

Now it is in the standard form:

$$\frac{dK}{dt} + P(t)K = Q(t),$$

with $P(t) = \delta$ (constant), $Q(t) = I$ (constant).

9.2 Integrating Factor Method (Step by Step)

We choose an **integrating factor** $\mu(t)$ such that:

$$\mu(t) = e^{\int P(t)dt} = e^{\int \delta dt} = e^{\delta t}.$$

Step 1: Multiply both sides of the differential equation by $\mu(t) = e^{\delta t}$:

$$e^{\delta t}\frac{dK}{dt} + \delta e^{\delta t} K = I e^{\delta t}.$$

Step 2: Recognize that the left-hand side is:

$$\frac{d}{dt}(e^{\delta t} K).$$

We can verify this:
Take $e^{\delta t}K$ and differentiate with respect to t:

$$\frac{d}{dt}(e^{\delta t} K) = e^{\delta t}\frac{dK}{dt} + \delta e^{\delta t} K.$$

So indeed:

$$\frac{d}{dt}(e^{\delta t} K) = I e^{\delta t}.$$

Step 3: Integrate both sides from 0 to t:

$$\int_0^t \frac{d}{ds}(e^{\delta s} K(s))ds = \int_0^t I e^{\delta s} ds.$$

Left-hand side:

$$\int_0^t \frac{d}{ds}(e^{\delta s} K(s))ds = e^{\delta t} K(t) - e^0 K(0) = e^{\delta t} K(t) - K(0).$$

Right-hand side:

$$\int_0^t I e^{\delta s} ds = I \int_0^t e^{\delta s} ds.$$

Compute:

$$\int_0^t e^{\delta s} ds = \left[\frac{1}{\delta} e^{\delta s}\right]_0^t = \frac{1}{\delta} e^{\delta t} - \frac{1}{\delta} e^0 = \frac{1}{\delta}(e^{\delta t} - 1).$$

Thus:

$$\int_0^t I e^{\delta s} ds = I \cdot \frac{1}{\delta}(e^{\delta t} - 1) = \frac{I}{\delta}(e^{\delta t} - 1).$$

Step 4: Put both sides together:

$$e^{\delta t}K(t) - K(0) = \frac{I}{\delta}(e^{\delta t} - 1).$$

Step 5: Solve for $K(t)$:

$$e^{\delta t}K(t) = K(0) + \frac{I}{\delta}(e^{\delta t} - 1).$$

Divide by $e^{\delta t}$:

$$K(t) = K(0)e^{-\delta t} + \frac{I}{\delta}(1 - e^{-\delta t}).$$

Capital Accumulation Solution

$$K(t) = K(0)e^{-\delta t} + \frac{I}{\delta}(1 - e^{-\delta t}).$$

- $K(0)e^{-\delta t}$ is the depreciating initial stock.
- $\frac{I}{\delta}(1 - e^{-\delta t})$ captures the build-up of new capital from investment.
- As $t \to \infty$, $e^{-\delta t} \to 0$, so

  $$K(\infty) = \frac{I}{\delta},$$

  which is the steady-state capital stock where net investment equals depreciation.

10. Debt Dynamics: Another Example

Let $B(t)$ be government debt at time t. Suppose:

$$\frac{dB}{dt} = rB + D(t),$$

where:

- r is the interest rate on debt.
- $D(t)$ is the (primary) deficit at time t.

10.1 Rearranging and Integrating Factor

Rewrite:

$$\frac{dB}{dt} - rB = D(t).$$

Here $P(t) = -r$, so:

$$\mu(t) = e^{\int -r dt} = e^{-rt}.$$

Multiply both sides by e^{-rt}:

$$e^{-rt}\frac{dB}{dt} - re^{-rt}B = D(t)e^{-rt}.$$

Left side is:

$$\frac{d}{dt}(e^{-rt}B).$$

Check:

$$\frac{d}{dt}(e^{-rt}B) = e^{-rt}\frac{dB}{dt} + (-re^{-rt})B = e^{-rt}\frac{dB}{dt} - re^{-rt}B.$$

Correct.

So we have:

$$\frac{d}{dt}(e^{-rt}B) = D(t)e^{-rt}.$$

Integrate from 0 to t:

$$\int_0^t \frac{d}{ds}(e^{-rs}B(s))ds = \int_0^t D(s)e^{-rs}ds.$$

Left-hand side:

$$e^{-rt}B(t) - e^0 B(0) = e^{-rt}B(t) - B(0).$$

Thus:

$$e^{-rt}B(t) - B(0) = \int_0^t D(s)e^{-rs}ds.$$

Solve for $B(t)$:

$$e^{-rt}B(t) = B(0) + \int_0^t D(s)e^{-rs}ds,$$

$$B(t) = e^{rt}\left[B(0) + \int_0^t D(s)e^{-rs}ds\right].$$

Debt Solution

$$B(t) = e^{rt}\left[B(0) + \int_0^t D(s)e^{-rs}ds\right].$$

Interpretation:

- $B(0)e^{rt}$ is the initial debt grown at interest rate r.
- The integral term represents the accumulation of discounted deficits over time.

11. Present Value as an Integral

In continuous time, if $Y(t)$ is a flow of income and r is the constant discount rate, the present value (PV) is:

$$PV = \int_0^\infty Y(t)e^{-rt}dt.$$

11.1 Example: Constant Income Stream

Suppose $Y(t) = Y$ (constant). Then:

$$PV = \int_0^\infty Ye^{-rt}dt = Y\int_0^\infty e^{-rt}dt.$$

Compute the integral:

$$\int_0^\infty e^{-rt}dt = \lim_{T\to\infty}\int_0^T e^{-rt}dt.$$

First compute up to T:

$$\int_0^T e^{-rt}dt = \left[-\frac{1}{r}e^{-rt}\right]_0^T = -\frac{1}{r}e^{-rT} + \frac{1}{r}e^0 = \frac{1}{r}(1 - e^{-rT}).$$

Now take the limit as $T \to \infty$:

$$\lim_{T\to\infty} \frac{1}{r}(1 - e^{-rT}) = \frac{1}{r}(1 - 0) = \frac{1}{r}.$$

So:

$$PV = Y \cdot \frac{1}{r} = \frac{Y}{r}.$$

This is the continuous-time analog of the perpetuity formula.

12. Summary and Economic Insights

Core Messages

- Integration is the mathematical tool that sums infinitely many tiny contributions (areas, flows).
- The definite integral is the limit of Riemann sums and gives areas under curves.
- The indefinite integral (antiderivative) reverses differentiation.
- In economics, integration is used to compute total cost, total revenue, consumer surplus, and more.
- A stock (capital, debt, knowledge) is often the integral of its net flow (investment, deficit, R&D).
- Differential equations model how economic variables evolve over continuous time.
- The integrating factor method allows us to solve a wide class of linear first-order differential equations.
- Present value in continuous time is an integral of discounted flows.